Md. Joynal Abedin Joy
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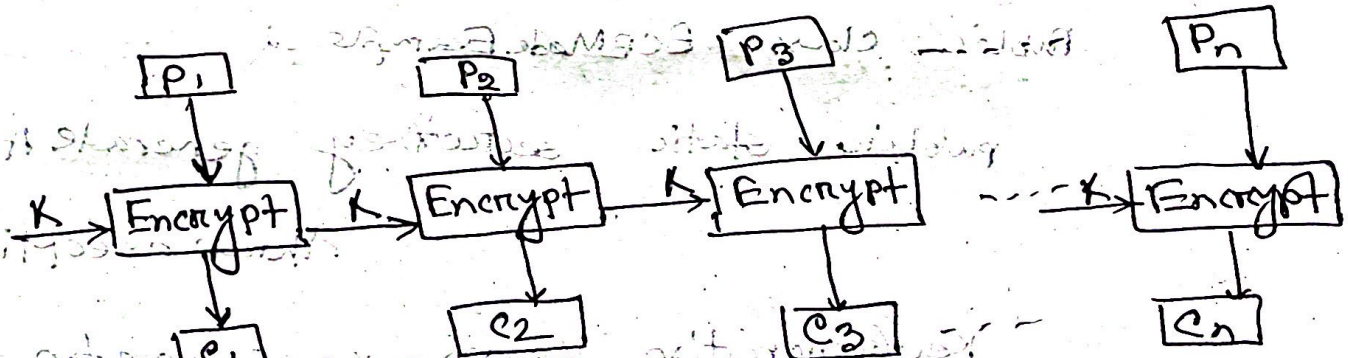
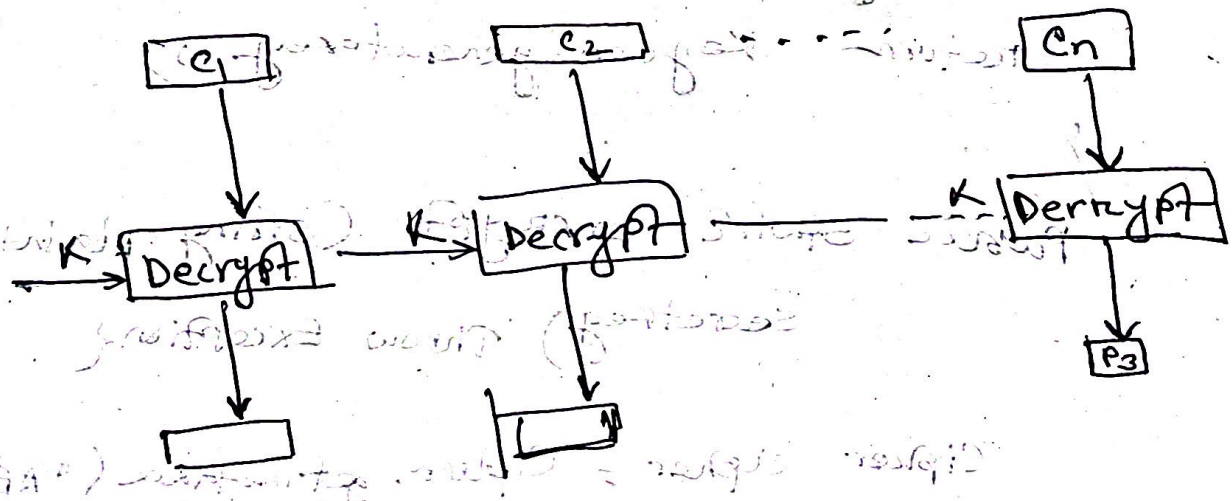
ECB (Electric codebook) : In an electric codebook each block of bits of plain text is encoded independently with the same key.

Block Diagram:





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Java implementation of ECB

```
import Javax.Crypto.Cipher;
import Javax.crypto.KeyGenerator;
Import Javax.crypto.secretKey;
import Javax.crypto.spec.secretKeyspace;
import Java.util.Base64;

Public class ECBModeExample2
    public static secretKey generateAESkey()
                                Throw exception{
    KeyGeneration Keygen = KeyGenerator.get
                        instance("AES");
        KeyGen.int(128);
    return KeyGen.generatekey();
}
Public static encrypt (string plaintext,
            secretkey) Throw Exception{
    Cipher cipher = Cipher.getInstance("AES/ECB/
                        PKCS5Padding");
```

```
Cipher.init(cipher.ENCRYPT_Mode secret);
byte[] encryptedBytes = cipher.doFinal(plaintext.get
                                        bytes());
return Base64.getEncoder().encodeString(EncryptBytes);
}
// Decrypt cipher text using AES in ECB mode
public static String decrypt(String encryptedText,
                             SecretKey secretKey) Throws Exception{
    Cipher cipher = Cipher.getInstance("AES/ECB/PCKS5");
    Cipher.init(Cipher.DECRYPT_MODE, secretKey);
byte // decryptedBytes = cipher.doFinal(Base64.get
                                        Decoder, decode(encryptedText));
    return new String(decryptedBytes);
public static void main(String[] args){
    try{ secret key = generateAESKey();
```

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```
String original = "Hello cyber world";
String encrypted = encrypt(original, key);
String decrypted = decrypt(encrypted, key);
System.out.println("original Text:" + original);
System.out.println("Encrypted text:" + encrypted);
System.out.println("Decrypted text:" + decrypted);
}
catch (Exception e) {
    e.printStackTrace();
}
}
```

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output:

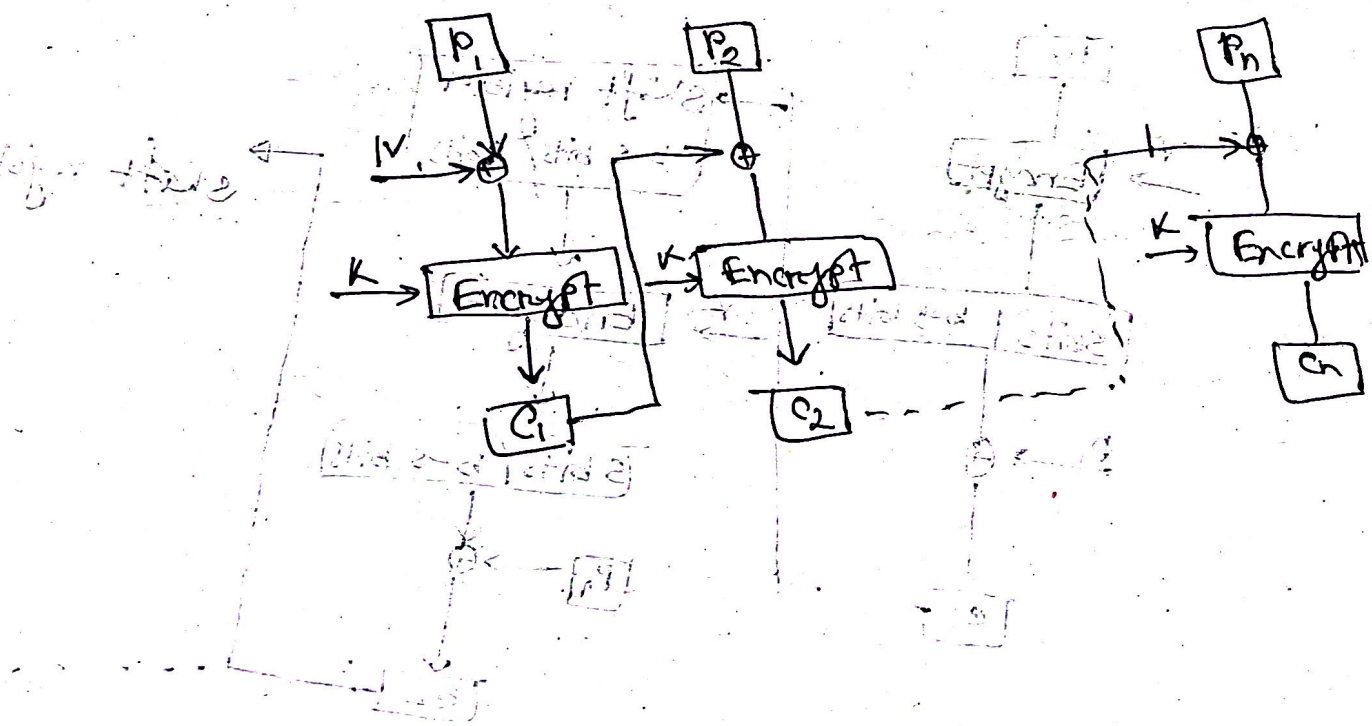
original Text : Hello cyber world !

Encrypted text: fRQt*yKBp 3AnX iV4IQWPIQRWRgU

Decrypted text: Hello cyber world !

CBC (Cipher Block chaining)

Encryption :



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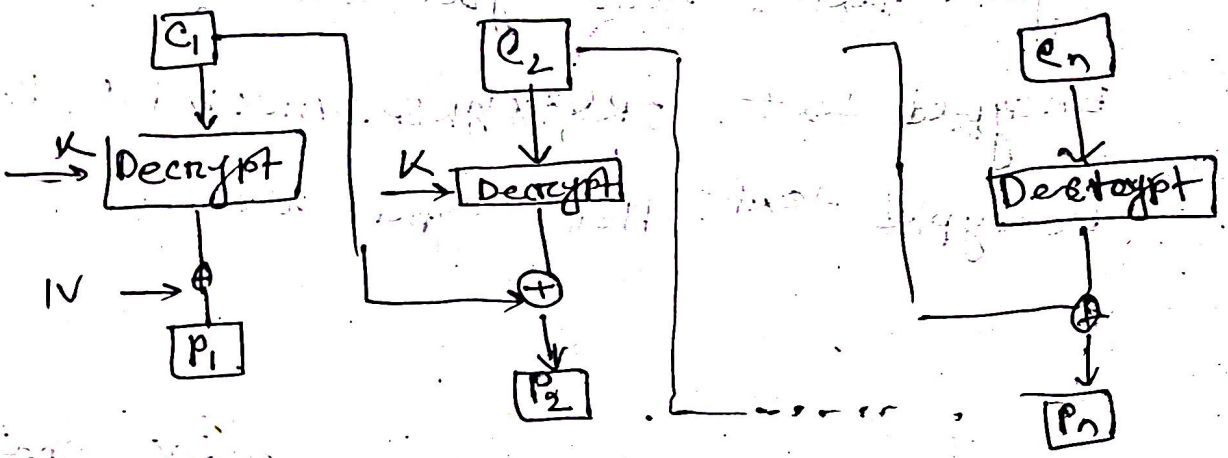
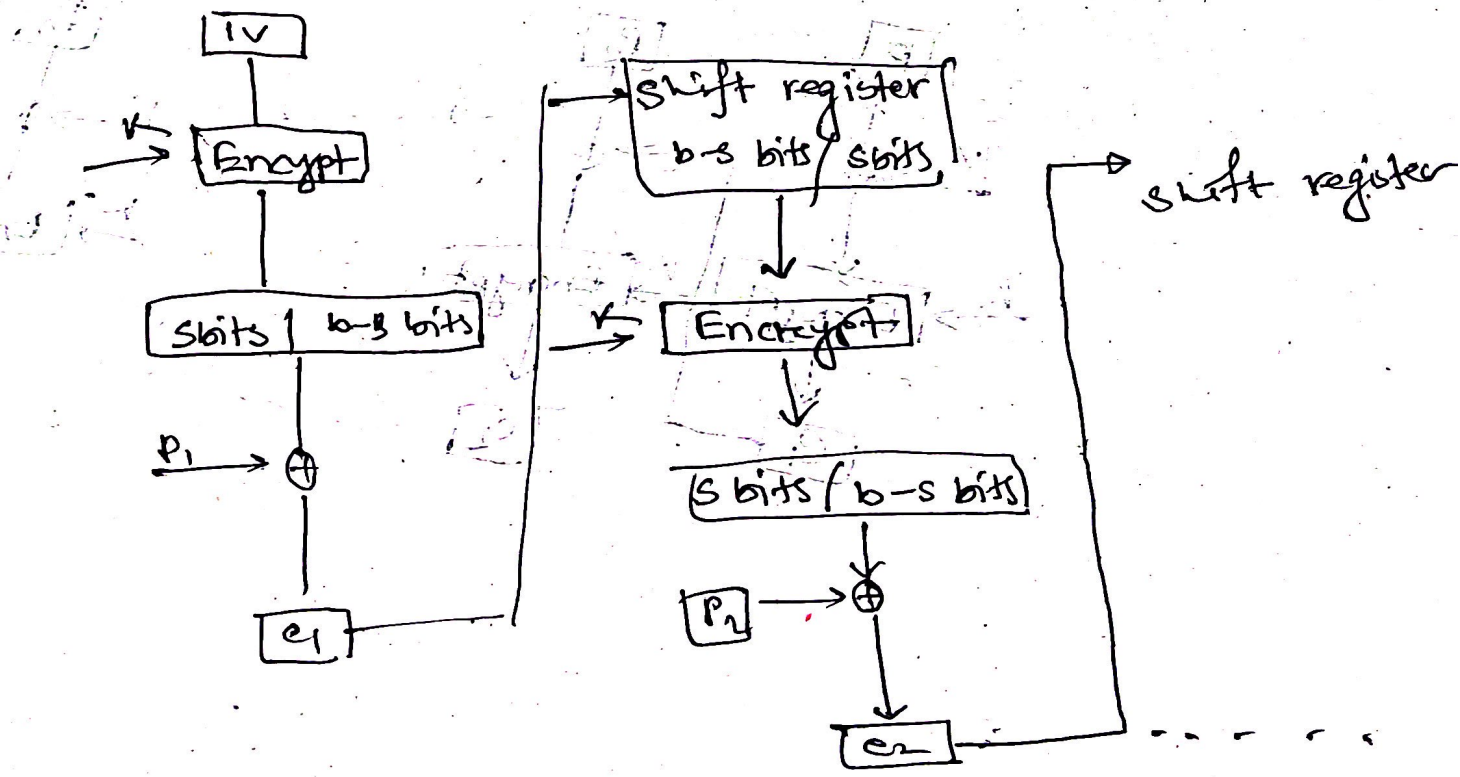
Decryption :



Cipher Feedback Mode (CFB)

Encryption



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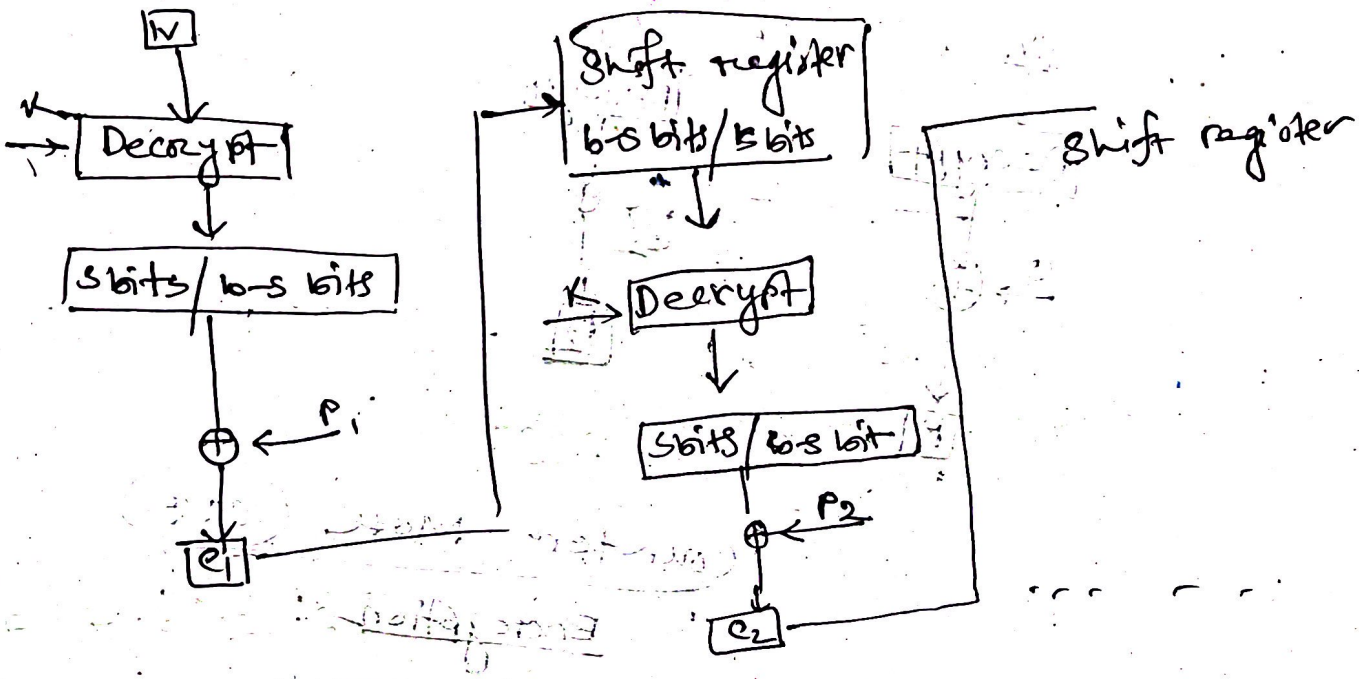
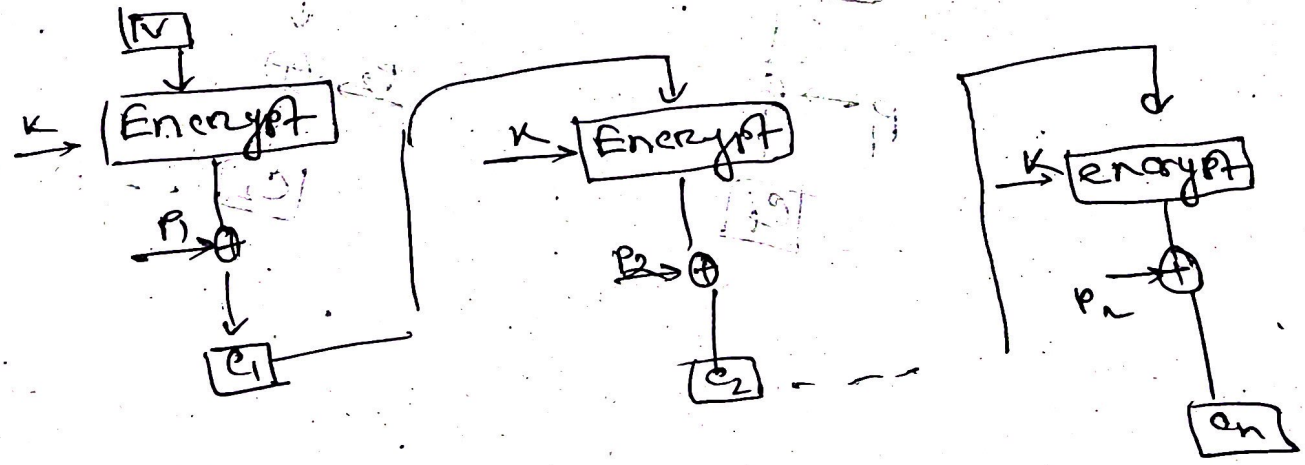
Decryption



Output feedback (OFB)

Encryption



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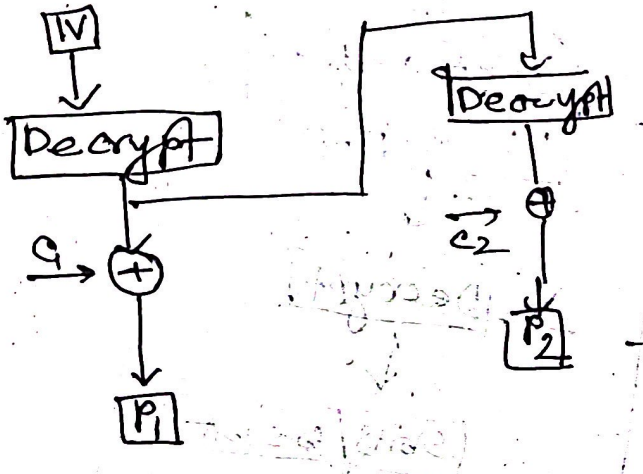
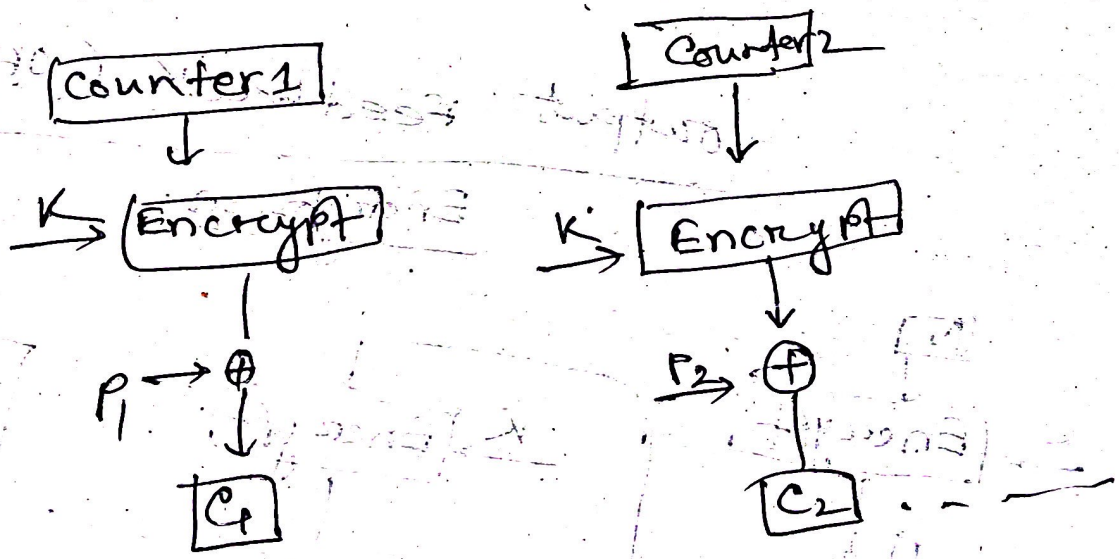
~~ECB~~

Decryption



Counter Mode (CTR)

Encryption :



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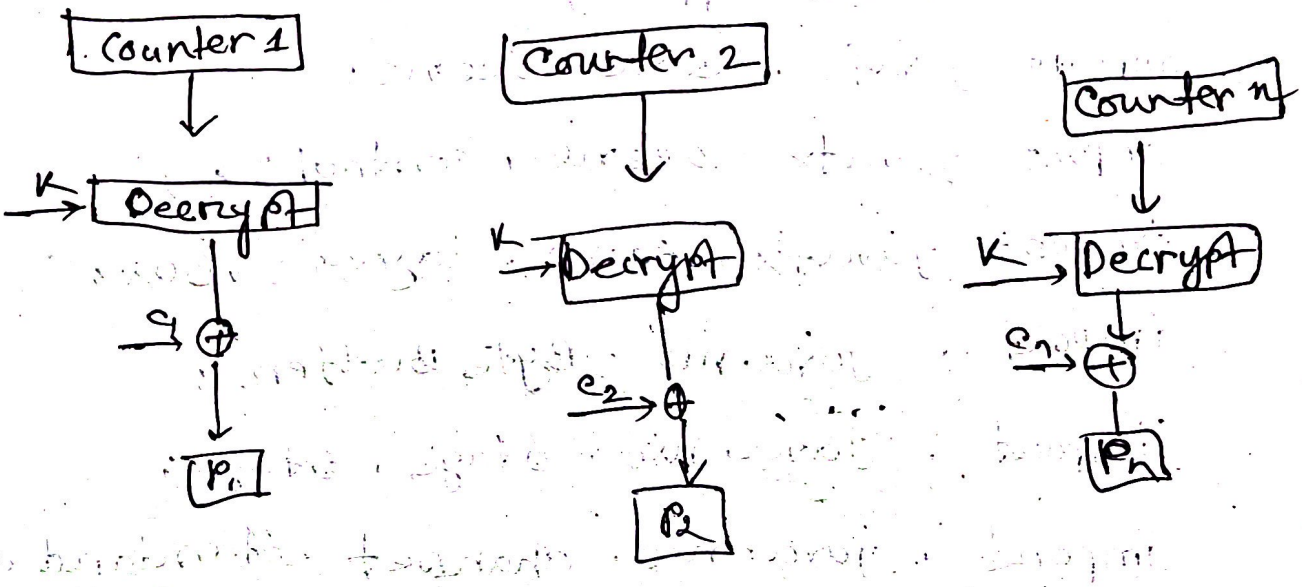
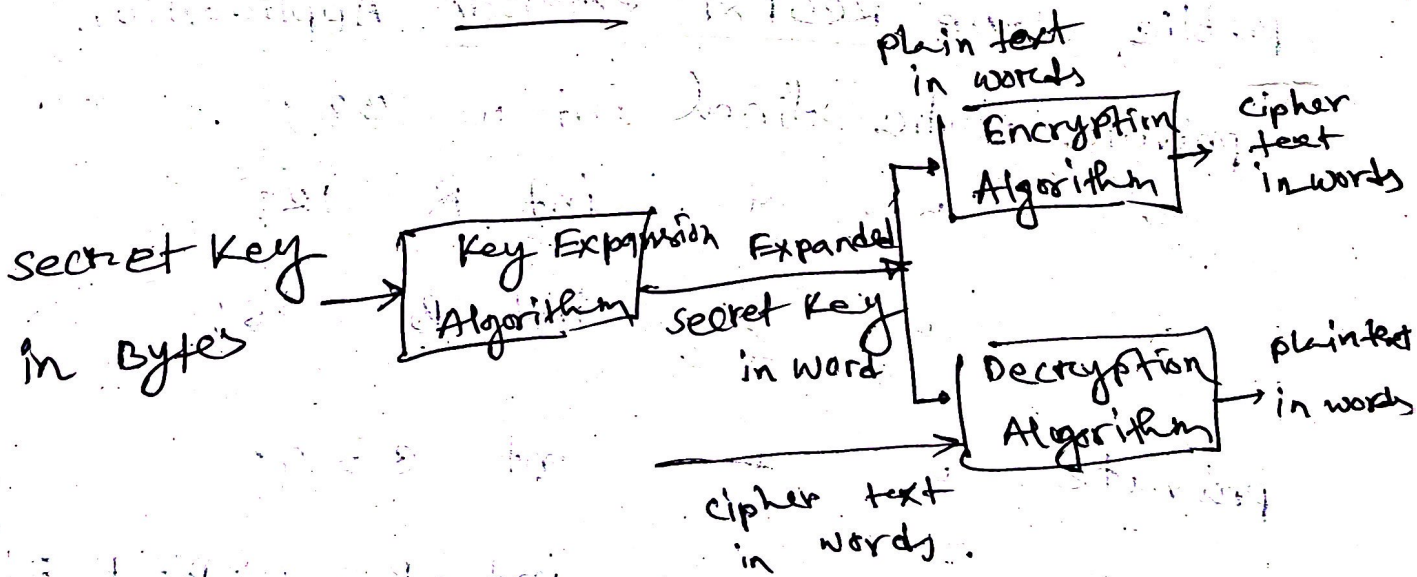
Decryption



RC5



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Code:

```
package org.example.rabfx;
import javafx.application.Application
import javafx.scene.Scene;
import javafx.scene.control.*;
import javafx.scene.layout.VBox;
import java.nio.ByteBuffer;
import java.nio.file.Files;
import java.nio.channels.StandardOpenOption;
import java.util.Arrays;

public class RC5Fx1 extends Application
    private static final int w = 32;
    "   "   "   int R = 12;
    "   "   "   int B = 16;
    private "   "   int C = 4;
    "   "   "   int P = 0xB7E15163;
    "   "   "   int Q = 0x9E3779B9;
    "   "   "   int rot(int x, int y) {
```

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```
    return (x<<y) | (x>>> (w-y));
}
Private static void rc5 key setup (byte [] key)
{
  private static void RC5 key setup (byte [] key)
    int [ ] = new int [c];
    for (int i = B-1; i>=0; i--) {
    L (i/4] = (L [i/4] <<8) + key [i] & 0xFF);
}
S [0] = P;
for (int i = 0; i <2 * (R+1) ; ++i){
      S [i] = S [i-1] + Q;
}
int A = 0, B = 0;
int i = 0, j = 0
Private static void rc5 Encrypt (int [] s, int [] data
{
    int A = data [0]
    int B = data [i]
```

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```
A = A + S[0];
B = B + S[1];
for (int i = 1; i <= R; ++i) {
    A = not | (A ^ B; B) + S[2*i];
    B = not | (B ^ A, A) + S[2*i+1];
}
data[0] = A;
data[1] = B;
}
Private static String encryptText (String plainText) {
    byte[] key = new byte[B];
    int[] S = new int[2 * (R*1)];
    RC5 key setup (key, 5);
byte[] plainBytes = plaintext.getBytes (standard
    charset UTF = 8);
```

```
while (buffer.hasRemaining()){
    int A = buffer.getInt();
    int B = buffer.getInt();
    int[] data = {A, B};
    rc5.Encrypt(s.data);
    ciphertex.append(String.format("%08x
            %08x", data[0], data[1]));
}
return ciphertext.toString().trim();
}
```

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```
@ overnide
  public void stunt (Stage . stage){
Textfield input = new Textfield ();
input . setpnomptText ("Enten Englinh to Encryp
Button enengptButton = new. Button ("Enengpt");
  TextAnea output = new . TextAnea ();
  output. set Edittable (false);
  outpat . set unapText (tnue);
enengptButton . set on Action (e → {
  string plaintext = input . getText ();
if (plaintext ! = null && ! plainText . is Empty ();
  string ciphenText = enengpt . Text (plainText)
}
else
  { output . setText ("Please enten nome text");
  }
}
```

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```
VBox layout = new VBox (10 input encryptButton
    layout.setStyle ("-fx-padding: 20;");

@Override
public void start (Stage stage){
Textfiel input = new Textfield ();
input.setpromptText ("Enter English to encry

Button enencypButton = new Button ("Encrypt")
  TextArea output = new TextArea ();
  output.setwraptext (true);
enencyptButton.setOnAction (e -> {
    String plainText = input.getText ();
if (plainText != null && ! plainText.isEmpty ()) {
  String ciphenText = enencypt.Text (plainTe
}
else
  {
  output.set.Text ("Please enter some text")
}
```

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```
VBox layout = new VBox (10, input encryptButton,
                        output);
layout.setStyle ("-fx-padding: 20;");
Scene scene = new Scene (layout, 500, 300);
Stage.setScence (scene);
Stage.show();
Public static void main (String[] args) {
        launch (args)
}
}
```

output

input : hello world

padding into 64 bit blocks

Block 1 : "hellowo" (8 bytes)

Block 2 : "rld10101010" (Padded)

Encrypted output

07c9f2a2, b1557b56, 24935031

original : hello world